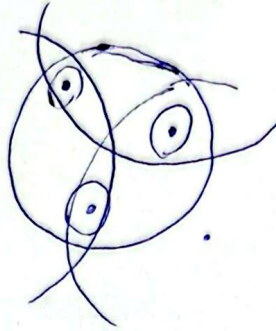
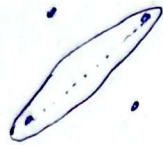


Q1)

1.1)



this shows that
a set of 3 points is
shattered



but in a 4 point setting, the
2 farthest points can't be
separated, therefore:

$$VCdim = 3$$

1.2) It's possible to show that a 7 point
set can be shattered Geometrically, but it's
very messy, so we won't do it here. we'll
show that an 8 point set.

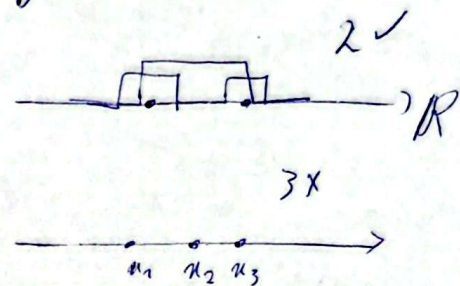
for this proof we assume that the points are
in a convex position, this doesn't reduce generalization
because even if we have interior points, it only
reduces the complexity. so we assume the
8 points, lie on the convex hull boundary.
if we order them cyclically

Q 1.2) continued:

because a line intersects a polygon in at most two edges, the "1.1.1.1" series can't be realized i.e.:

there's no triangle that captures these three points.
four

1.3) we'll prove this using induction

base case $n=1 \rightarrow$ 
(can't be realized $\{x_1, x_3\}$)

now imagine this fact holds for $\mathbb{R}^{n-1}$

we add a dimension, and two points

we can realize every subset within $\mathbb{R}^{n-1}$, now we have to

show that for $\forall C \subseteq X$ $C \cup \{x_3\}$, $C \cup \{x_1, x_2\}$, $C \cup \{x_2\}$

also holds. it's like the one dimensional

case since we already know how to control the Hyper Cubes parameters: $a_i \leq y_i \leq b_i$ for $1 \leq i \leq n-1$

we just have to handle for $i=n$.

$\rightarrow C \cup \{x_1\} \rightarrow x_1 - \frac{a}{2} \leq y_i \leq x_1 + \frac{a}{2}$, ... (and like wise for $\{x_2\}$, $\{x_1, x_2\}$)

it's obvious that adding another point will ruin the argument

(Q2)

2.1) we'll show that every α -cover w.r.t J_1

is also an α -cover for J_p norm given that $1 \leq p \leq \infty$

first note that for the empirical measure

$$\mu_n := \frac{1}{n} \sum_K \delta_{x_n} \quad \left(\frac{1}{n} \sum |f(x_i) - v_i|^p \right)^{1/p} = \|f - v\|_{L^p(\mu_n)}$$

therefore for $\forall f \in \mathcal{F}, \exists v \in \mathcal{V}$ s.t.

$$\|f - v\|_{L^1(\mu_n)} \leq \alpha \quad (\text{given that } \mathcal{V} \text{ is an } \alpha\text{-cover})$$

since the L^p norm is monotone, we have

$$\|f - v\|_{L^p(\mu_n)} \leq \|f - v\|_{L^1(\mu_n)} \leq \alpha$$

therefore every α -cover on J^1 is also an α -cover

for J^p . so

$$N_p(\mathcal{F}, \alpha, x^n) \leq N_1(\mathcal{F}, \alpha, x^n)$$

□.

2.2) the desired inequality follows from a simple 1-level discretization argument.

$$\hat{R}(F) = \mathbb{E}_{\varepsilon} \left[\sup_{f \in F} \frac{1}{n} \sum \varepsilon_i f(x_i) \right]$$

$$\frac{1}{n} \sum \varepsilon_i f(x_i) = Z_f$$

first we'll show that every Z_f is sub-gaussian

from Bernstein Lemma we know that $\sum \alpha_i x_i$ is $(\sum \alpha_i^2) \sigma^2$ sub-g, ε is 1-subg.

therefore Z_f is $\frac{\sum f(x_i)^2}{n^2}$ - subg

also note that $\frac{\sum f(x_i)^2}{n^2} \leq \frac{1}{n} \quad (\|f\|_{\infty} \leq 1)$

now we'll show that this Process is Lipschitz

$$Z_f - Z_g = \frac{1}{n} \sum \varepsilon_i (f(x_i) - g(x_i))$$

$$\leq \frac{1}{n} \sum |f(x_i) - g(x_i)| = \|f - g\|_{L^1(\mu_n)}$$

therefore Z_f is a 1-Lipschitz Process w.r.t

$\|\cdot\|_{L^1(\mu_n)}$ (norm)

2.2) continued:

Lipschitz maximal inequality: (lemma 5.7 Van Handel)

$$\mathbb{E}[\sup_t X_t] \leq \inf_{\varepsilon} \left\{ \varepsilon \mathbb{E}(C) + \sqrt{2\sigma^2 \log(N(\tau, d, \varepsilon))} \right\}$$

for our case: $C = 1$, $\varepsilon = \alpha$, $\sigma^2 = \frac{1}{n}$

$$\rightarrow \mathbb{E}[\sup_t Z_t] = \hat{R}(F) \leq \inf_{\alpha} \left\{ \alpha + \sqrt{\frac{2 \log N_1(F, \alpha, n)}{n}} \right\}$$

□.

2.3) first, let's calculate the $(\alpha$ -Cover) N w.r.t. $L_1(\mu_n)$ norm.

let's say that g is in the α -Cover G for F .

$$\|f - g\|_{L_1(\mu_n)} = \frac{1}{n} \sum \langle w - h, x_i \rangle$$

$$\langle w - h, x_i \rangle \leq \|w - h\|_{p^*} \|x_i\|_p \leq \|w - h\|_{p^*}$$

$$\rightarrow \|f - g\|_{L_1(\mu_n)} \leq \|w - h\|_{p^*} < \alpha \rightarrow \text{therefore for an } \alpha\text{-cover on } F$$

we need to cover $[-1, 1]^d$ w.r.t l^∞ norm, a simple grid argument, gives the upper bound:

$$N_1(F, \alpha, n) \leq \left\lceil \frac{2}{\alpha} \right\rceil^d$$

now that we've achieved our upper bound on N
we can bound $\hat{R}(F)$.

$$\hat{R}(F) \leq \inf_{\alpha > 0} \left\{ \alpha + \sqrt{\frac{2 \log(N(\dots))}{n}} \right\}$$

we'll now minimize the RHS.

$$N(\dots) \approx \left(\frac{1}{\alpha}\right)^d \rightarrow \text{RHS} = \alpha + \sqrt{\frac{2d \log \frac{1}{\alpha}}{n}}$$

$$\rightarrow \frac{d}{d\alpha}(\text{RHS}) = 1 + \frac{\sqrt{\frac{2d}{n}}}{2\sqrt{\log \frac{1}{\alpha}}} = 0$$

$$\rightarrow \alpha^* \asymp \sqrt{\frac{d}{n}} \frac{1}{\sqrt{\log(n/d)}}$$

$$\rightarrow \text{RHS}|_{\alpha=\alpha^*} = \mathcal{O}\left(\sqrt{\frac{d \log n}{n}}\right)$$

2.4) first let's calculate that our Process
is sub-gaussian wrt what norm

$$\rightarrow Z_f - Z_g = \frac{1}{n} \sum \mathcal{E}_i(f(x_i) - g(x_i))$$

from Bern stein lemma we know that

$Z_f - Z_g$ is a subgaussian random variable.

2.4) continued:

so we've established that $Z_f - Z_g$ is subg
with variance proxy: $\frac{\sum (f(x_i) - g(x_i))^2}{n^2} \rightarrow$

$\rightarrow = \frac{\|f - g\|_{L^2(\mu_n)}^2}{n}$, therefore this Process is subg

wrt the empirical norm: $\frac{\|\cdot\|_{L^2(\mu_n)}}{\sqrt{n}}$

also it was established in 2.2:

$$Z_f - Z_g \leq \|f - g\|_{L^2(\mu_n)} \leq \|f - g\|_{L^2(\mu)} = \sqrt{n} \times \frac{\|f - g\|_{L^2(\mu)}}{\sqrt{n}}$$

therefore our Process is $\sqrt{n}$ -Lipschitz wrt $\frac{\|\cdot\|_{L^2(\mu_n)}}{\sqrt{n}}$ norm
with the basis covered, we'll now move on to proving
the inequality.

from Pset 3 (Q2) and Van-Vander Vaart (5.31)

we have:

$$\mathbb{E} \left[\sup_{t \in T} X_t \right] \leq \inf_{\delta > 0} \left\{ 2 \delta \mathbb{E}[C] + 12 \int_{\delta}^{\infty} \sqrt{\log(N(\tau, \delta, \varepsilon))} d\varepsilon \right\}$$

(the proof will be added in the appendix)

2.4) continued:

if we apply our parameters to the improved chaining we'd have:

$$\hat{R}(\bar{F}) \leq \inf_{\alpha > 0} \left\{ 2\sqrt{n} \alpha + 12 \int_{\alpha}^{\infty} \log(\mathcal{N}(\bar{F}, \frac{\|\cdot\|_{L^2(\mu_n)}}{\sqrt{n}}, \delta)) d\delta \right\}$$

$$\rightarrow \mathcal{N}(\bar{F}, \frac{\|\cdot\|_{L^2(\mu_n)}}{\sqrt{n}}, \delta) = \mathcal{N}(\bar{F}, \|\cdot\|_{L^2(\mu_n)}, \delta\sqrt{n})$$

$$\rightarrow \int_{\alpha}^{\infty} \log(\dots) d\delta \rightarrow \frac{1}{\sqrt{n}} \int_{\alpha\sqrt{n}}^1 \log(\mathcal{N}(\bar{F}, \|\cdot\|_{L^2(\mu_n)}, \delta')) d\delta'$$
$$\delta' = \delta\sqrt{n}$$

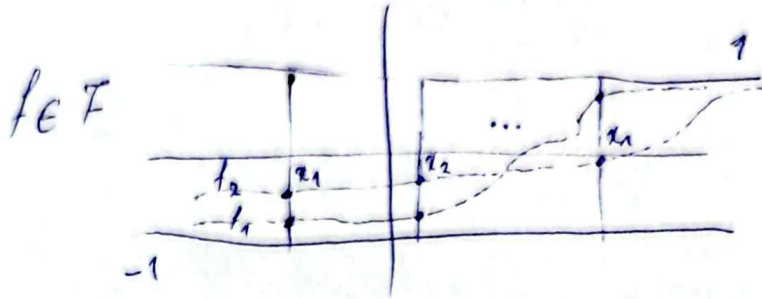
$\rightarrow \alpha' = \sqrt{n} \times \alpha$, minimizing on α, α' has the same result:

$$\hat{R}(\bar{F}) \leq \inf_{\alpha' > 0} \left\{ 2\alpha' + \frac{12}{\sqrt{n}} \int_{\alpha'}^1 \log(\mathcal{N}(\dots)) d\delta' \right\}$$

2.5) the correct form to that bound is:

$$N_\infty(F, \alpha, n) \leq \left\lceil \frac{2}{\alpha} \right\rceil^n$$

this bound is evident from a simple argument.



in order to achieve a distance $\|f - g\|_{L^\infty(x_n)} < \alpha$

we have to have functions α -apart at each sample, since we have to cover a distance of 2.

vertically we'll need $\left\lceil \frac{2}{\alpha} \right\rceil$ functions

(and since we have n points, we'll need a total

of $\left\lceil \frac{2}{\alpha} \right\rceil^n$.

applying this bound to part 2.2 yields:

$$\hat{R}_n(F) \leq \inf \left\{ \alpha + \sqrt{\frac{2 \log N_\infty(\dots)}{n}} \right\}$$

$$\leq \inf \left\{ \alpha + \sqrt{2 \log \left(\frac{2}{\alpha} \right)} \right\} = O(1)$$

this is a terrible result since it doesn't decay with sample size.

2.3) continued

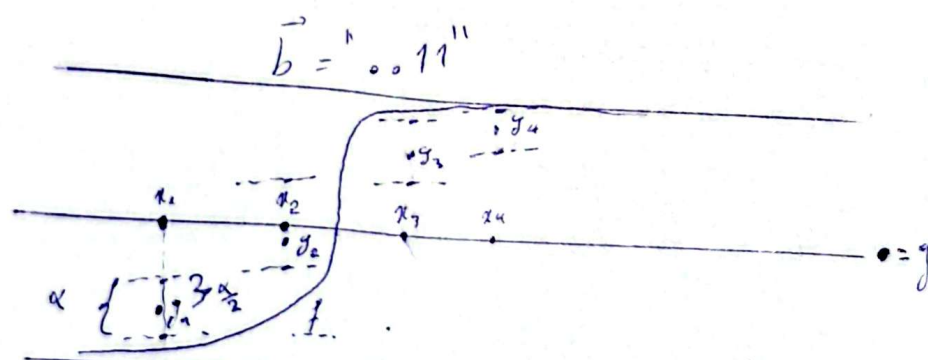
applying this bound to 2.4 yields:

$$\hat{R}(T) \leq \inf \left\{ 4\alpha + \frac{12}{\sqrt{n}} \int_{\alpha}^1 \sqrt{f_2(x)} dx \right\}$$

$$\leq \inf \left\{ 4\alpha + 12 \int_{\alpha}^1 \sqrt{f_2\left(\frac{1}{8}\right)} dx \right\} = O(1)$$

this result is as weak as part 2.2.

2.6) to explain the Intuition behind the Fat dim of a class of functions, it's fair to say that it's a real valued extension of the VC-dimension and classical shattering, instead of binary valued functions, we now have a threshold band, determining the binary values let's explain visually:



in order to be able to capture binary strings like "010"
the spacing between g_i 's should be in a certain way

2.7) as it was shown in the last part,

in order to capture binary strings

like: $\underset{j}{1} \underset{j+1}{0}$ the above inequality

has to be satisfied: $g_{j+1} - \frac{\alpha}{2} > g_j + \frac{\alpha}{2}$

therefore $g_{j+1} - g_j > \alpha$. this indicates

that $[-1, 1]$ can be divided into at most

$\lceil \frac{2}{\alpha} \rceil$ sections of length α , this argument

gives us an upper bound on $N(F, \alpha)$.

therefore we have: $\text{Fatdim}(F, \alpha) \leq \lceil \frac{2}{\alpha} \rceil < 4/\alpha$

2.8) *

2.9) from 2.4: $\hat{R}(F) \leq \inf_{\alpha > 0} \left\{ 4\alpha + \frac{12}{\sqrt{n}} \int_{\alpha}^1 \sqrt{\log N_2(F, \delta, \alpha^n)} d\delta \right\}$

from 2.8 we have: $N_2(F, \alpha, \alpha^n) \leq \left(\frac{2}{\alpha}\right)^{K \cdot \text{Fatdim}(F, \alpha)}$

$$\rightarrow \hat{R}(F) \leq \inf_{\alpha} \left\{ 4\alpha + \frac{12}{\sqrt{n}} \int_{\alpha}^1 \sqrt{K \cdot \text{Fatdim}(F, \delta) \cdot \log \left(\frac{2}{\delta}\right)} d\delta \right\}$$

Q3) first we'll show there's a Correspondence between the Concept of shattering between Classes of sets and a Class of Binary Functions.

shattering definition for Classes of sets:

$$C \subseteq 2^X, \quad S \subseteq X, \quad |S| = m < \infty$$

we'll say S is shattered by C if: $C \cap S = 2^S$.

now Consider this definition:

$$\mathcal{H}: \text{class of functions}; \quad X \rightarrow \{0,1\}, \quad |S| \leq m$$
$$S = \{x_1, \dots, x_m\}$$

$$\text{it: } \forall b \in \{0,1\}^m, \exists h \in \mathcal{H} \text{ s.t. : } b = (h(x_1), \dots, h(x_m))$$

then S is shattered by $\mathcal{H}$.

the Correspondence Comes from the fact that abinary function is essentially the same as the logical operator

(ϵ). therefore if we show the above equation for $\forall b \in \{0,1\}^m$

we've essentially built the power set of S .

with the result above now i'll argue that $VC \dim(\mathcal{H})$
 $= \mathcal{H} \dim(\mathcal{H})$

in order to show this it's enough to show that

the Concept is the same for both in case of $k=2$

Q3) 3.1) continued:

note that in the def. in the psec. ($\mathcal{C} \equiv \mathcal{S}$) (i.e. $\mathcal{C}$ plays the role of $\mathcal{S}$), without loss of generality we can

assume: $f_0(x) = 0, f_1(x) = 1$ (since: $\forall x \in X: f_0(x) \neq f_1(x)$)

$\forall B \in \mathcal{C}$ is $B = \{b_1, \dots, b_j\}$ (random), $\mathcal{C} = \{b_1, \dots, b_m\}$

$\rightarrow (h(x_1), \dots, h(x_m)) = (\cdot \cdot \cdot) \rightarrow$ this array is the
therefore iterating over all subsets $\left(\begin{array}{l} \text{same as the binary} \\ \text{string that we had} \\ \text{before.} \end{array} \right)$

of $\mathcal{C}$ yields the same result of shattering for VC.

therefore: $VC \dim(\mathcal{H}) = N \dim(\mathcal{H})$

3.2) we're gonna use a proof by induction on d, n .

first we'll choose an arbitrary dimension point

$x_0 \in X$. define: $X' := X \setminus \{x_0\}$

$\mathcal{H}|_{X'} := \{h|_{X'} : h \in \mathcal{H}\}$ (restrict the class to X')

$\mathcal{H}_{ij} := \left\{ f: X' \rightarrow [K] \mid \begin{array}{l} \exists h \in \mathcal{H} \text{ s.t. } h|_{X'} = f, h(x_0) = i \\ \wedge \\ \exists h' \in \mathcal{H} \text{ s.t. } h'|_{X'} = f, h'(x_0) = j \end{array} \right\}$

Q 3) 3.2 continued:

we want to show: $|\mathcal{H}| \leq |\mathcal{H}|_{X'} + \sum_{i,j} |\mathcal{H}_{ij}|$

in order to show the inequality above, holds, we'll define

$$L(f) := \{f \in \mathcal{H} : \exists h \in \mathcal{H} \text{ s.t. } h|_{X'} = f, h(x_i) = 1\}$$

then by definition we have: $|\mathcal{H}| = \sum_{f \in \mathcal{H}|_{X'}} |L(f)|$

now note that if $|L(f)| = 1$, f contributes 1 hypothesis.

and if $|L(f)| \geq 2$ there is $f \in \mathcal{H}_{ij}$ (for some i, j)

therefore we can bound $|L(f)| \leq 1 + \sum_{i,j} 1 \{f \in \mathcal{H}_{ij}\}$

summing over all $f \in \mathcal{H}|_{X'}$ gives:

$$|\mathcal{H}| \leq |\mathcal{H}|_{X'} + \sum_{i,j} |\mathcal{H}_{ij}|. \quad \text{now note that } \text{Ndim}(\mathcal{H}|_{X'}) \leq \dots$$

$\text{Ndim}(\mathcal{H}) = d$. (this is showable by the definition of Ndim)

by the induction base: $|\mathcal{H}|_{X'} \leq (M-1)k^{2d}$

now we'll show that $\text{Ndim}(\mathcal{H}_{ij}) \leq d-1$.

we'll prove this by contradiction, say there exists a set C that shatters $\mathcal{H}_{ij}$ with $|C| = d$.

now by the extension property of $\mathcal{H}_{ij}$ we know

that there are two functions f_0, f_1 such that...

Q 3.2) continued:

such that $j_1(x_i) = i, j_2(x_i) = j$ and $j_1|_C = f_0, j_2|_C = f_1$

therefore every subset of $C \cup \{x\}$ is realizable

this contradicts with the fact that $\text{Ndim}(\mathcal{H}) = d$, because we just realized a set of size $d+1$.
(shattered)

therefore $\text{Ndim}(\mathcal{H}_{ij}) \leq d-1 \rightarrow |\mathcal{H}_{ij}| \leq (n-1)^{d-1} K^{2d-2}$

$$\rightarrow |\mathcal{H}| \leq (n-1)^d K^{2d} + K^2 (n-1)^{d-1} K^{2d-2}$$

$$\leq K^{2d} (n-1)^{d-1} (n-1+1) \leq n^d K^{2d}. \quad \square.$$

Q 4) $P(\lambda_{\max}(S_n) > \delta) \leq P(\phi(\lambda_{\max}(S_n)t) > \phi(t\delta))$
($t > 0$)

4.1)

$$\leq P(\lambda_{\max}(\phi(tS_n)) > \phi(t\delta)) \leq \frac{E[\lambda_{\max}(\phi(tS_n))]}{\phi(t\delta)}$$

(since $\phi(t)$ is monotone and pos for $t > 0$)

$\rightarrow$ now note that $\lambda_i > 0$

for $\phi(tS_n) \rightarrow \lambda_{\max}(\phi(tS_n)) \leq \text{tr}[\phi(tS_n)]$

$$\rightarrow \leq \frac{E[\text{tr}(\phi(tS_n))]}{\phi(t\delta)} = \frac{\text{tr}[E\phi(tS_n)]}{\phi(t\delta)} \quad \left(\begin{array}{l} \text{tr and } E \\ \text{are linear operators} \end{array} \right)$$

4.2) $e^{tA} = I + tA + (e^{tA} - tA - I)$

define: $f(x) = \frac{e^{tx} - tx - 1}{x^2}$

$\rightarrow e^{tA} = I + tA + A \cdot f(A) \cdot A$

Q 4.2) continued: since $f(X)$ is monotone and $\|A\| < 1$

we have $f(A) \leq f(1) \cdot I = \varphi(1) \cdot I$ ($f(u) = \varphi(u)$)

$$\rightarrow e^{tA} \leq I + tA + \varphi(t)A^2 \quad \mathbb{E} e^{tA} \leq I + \varphi(t)\mathbb{E}A^2$$

$$\leq e^{\varphi(t)\text{Var}(A)} \quad (I + X \leq e^X)$$

$$\rightarrow \log \mathbb{E} e^{tA} \leq \varphi(t) \text{Var}(A)$$

Q 4.3) $\mathbb{P}[\lambda_{\max}(S_n) > \delta] \leq \frac{\text{Tr}\{ \mathbb{E}[\varphi(tS_n)] \}}{\varphi(t\delta)}$

$$\varphi(tS_n) = e^{tS_n} - tS_n - I \rightarrow \mathbb{E}[\varphi(tS_n)] = \mathbb{E}[e^{tS_n}] - I$$

$$\rightarrow \dots \leq \frac{\text{Tr}\{ \mathbb{E} e^{tS_n} \} - d}{\varphi(t\delta)} \leq \frac{\text{Tr}\{ \exp(\sum \log \mathbb{E} e^{tA_k}) \} - d}{\varphi(t\delta)}$$

(this follows from Lieb's theorem), $\log \mathbb{E} \exp tA_k \leq \varphi(t) \text{Var}(A_k)$

$$\rightarrow \leq \frac{\text{Tr} \left[e^{\varphi(t) \sum \text{Var}(A_k)} \right] - d}{\varphi(t\delta)} \neq \frac{\text{Tr} [e^{\varphi(t)V}] - d}{\varphi(t\delta)}$$

$$\text{Tr} [e^{\varphi(t)V}] = \sum e^{\varphi(t)\lambda_i} \quad (\text{since } e^{\varphi(t)x} \text{ is convex we have:})$$

$$e^{\varphi(t)\lambda_i} \leq \left(1 - \frac{\lambda_i}{\|V\|}\right) e^{\varphi(t)\lambda_0} + \frac{\lambda_i}{\|V\|} e^{\varphi(t)\|V\|}$$

$$\text{summing over all } i \text{ yields: } \text{Tr} [e^{\varphi(t)V}] \leq d + \frac{\text{Tr}(V)}{\|V\|_1} e^{\varphi(t)\|V\|}$$

Q 4.3) continued:

$$\dots \frac{\frac{\text{Tr}(V)}{\|V\|_F} e^{\varphi(t)\|V\|_F} + d - d}{\varphi(t\delta)} \quad \square.$$

minimizing on "t" yields the desired result.

4.4) in order to provide a bound on the norm of the sum of the vectors, we'll use Hermitian dilation trick!

$$A_K = \mathcal{H}_\phi(X_K) = \begin{pmatrix} 0 & X_K^T \\ X_K & 0 \end{pmatrix} \quad \left(\begin{array}{l} \text{note that eigen values} \\ \text{of } A_K \text{ are the same} \\ \text{as } X_K \text{ with } \pm \text{ signs} \end{array} \right)$$

$$A_K \rightarrow \text{rank } 2 \rightarrow \text{eigen values: } \pm \|X_K\|_2 \text{ and eigenvectors: } \begin{pmatrix} \pm \frac{X_K}{\|X_K\|_2} \end{pmatrix}$$

$$\rightarrow \mathcal{P}(\|\sum X_K\|_2 > \delta) = \mathcal{P}(\lambda_{\max}(\sum A_K) > \delta)$$

$$\text{Var}(A_K) = E[A_K A_K^T] = E\left[\begin{pmatrix} X_K^T X_K & 0 \\ 0 & X_K X_K^T \end{pmatrix}\right]$$

$$= \begin{pmatrix} \text{Tr}(C) & 0 \\ 0 & C \end{pmatrix} \rightarrow V = \sum \text{Var}(A_K) = n \begin{pmatrix} \text{Tr}(C) & 0 \\ 0 & C \end{pmatrix}$$

$$\rightarrow \text{Tr}(V) = n \cdot 2 \cdot \text{Tr}(C), \quad \|V\|_{op} = n \max\{\text{Tr}(C), \|C\|_{op}\}$$

$$\text{since } C \succ 0 \rightarrow \|V\|_{op} = n \text{Tr}(C)$$

Q 4.4) continued:

$$\rightarrow P(\dots) \leq 2 \int_1^\infty \left\{ \frac{\exp(\psi(t) \cdot n \cdot \text{Tr}(C))}{\psi(t\delta)} \right\}$$

$\rightarrow$ minimize on t :

$$\leq 2 \exp \left(\frac{-\delta^2}{2(n \text{Tr}(C) + \delta/3)} \right)$$

this bound goes to zero for $\delta \approx O(\sqrt{n})$

let's write this as Bernstein type:

$$P(\lambda_{\max}(\sum A_k) > \delta) \leq d e^{-\frac{\delta^2}{2\sigma^2 n}}$$

$$\sigma^2 = \left\| \frac{1}{n} \sum V_i \right\|_2$$

because our A_k are bounded with one then they are subgaussian with variance proxy matrix I .

$$\rightarrow \sigma^2 = 1$$

$$\rightarrow P(\dots) \leq d \exp\left(-\frac{\delta^2}{2n}\right) \rightarrow \delta = O(\sqrt{n \log d})$$

using Bernstein has the bad estimate of $\text{rank}(A) \leq d$ which adds the constant $\sqrt{\log d}$, we get rid of this constant when we use $\text{srnk}\left(\frac{\text{Tr}(V)}{\|V\|_{\text{F}}}\right)$ instead of regular rank.

Q5) 5.1)

$$(e^{\theta \lambda_{\max}(Y)}) = \lambda_{\max}(e^{\theta Y})$$

$$\mathcal{P}(\lambda_{\max}(Y) > t) \stackrel{(\theta > 0)}{=} \mathcal{P}(e^{\theta \lambda_{\max}(Y)} > e^{\theta t}) = \mathcal{P}(\lambda_{\max}(e^{\theta Y}) > e^{\theta t})$$

$$\leq e^{-\theta t} \mathbb{E}[\lambda_{\max}(e^{\theta Y})] \leq e^{-\theta t} \mathbb{E}[\text{tr}(e^{\theta Y})] \quad \left(\begin{array}{l} \text{minimizing on } \theta \\ \text{yields the desired} \\ \text{result} \end{array} \right) \quad \square.$$

(because $e^{\theta Y} \succeq 0$)

$$\rightarrow \mathcal{P}(\lambda_{\min}(Y) < t) \stackrel{\theta < 0}{=} \mathcal{P}(e^{\theta \lambda_{\min}(Y)} > e^{\theta t}) = \mathcal{P}(\lambda_{\min}(e^{\theta Y}) > e^{\theta t})$$

...
the rest is the same.

because $e^{\theta x}$ ($\theta < 0$) is a decreasing function we have: $e^{\theta \lambda_{\min}(Y)} = \lambda_{\min}(e^{\theta Y})$

5.2)

$\theta > 0$

$$\begin{aligned} \mathbb{E}[\lambda_{\max}(Y)] &= \frac{1}{\theta} \mathbb{E}[\log(\exp(\lambda_{\max}(\theta Y)))] \leq \frac{1}{\theta} \log \mathbb{E}[\exp(\theta \lambda_{\max}(Y))] \\ &= \frac{1}{\theta} \log \mathbb{E}[\lambda_{\max}(e^{\theta Y})] \leq \frac{1}{\theta} \log \mathbb{E}[\text{tr}(e^{\theta Y})] \quad \left(\begin{array}{l} \text{jensen} \\ \log \text{ is Concave} \end{array} \right) \end{aligned} \quad \square.$$

$$\mathbb{E}[\lambda_{\min}(Y)] \stackrel{(\theta < 0)}{=} \frac{1}{\theta} \mathbb{E}[\lambda_{\max}(\theta Y)] = \frac{1}{\theta} \mathbb{E}[\log(\exp(\lambda_{\max}(\theta Y)))]$$

$$\stackrel{\text{jensen}}{\geq} \frac{1}{\theta} \log \mathbb{E}[\exp(\lambda_{\max}(\theta Y))] \geq \frac{1}{\theta} \log \mathbb{E}[\text{Tr}(e^{\theta Y})] \quad \square.$$

optimizing on θ yields the desired result in each part. note that because $\theta < 0$, the inequalities flip in the second part.

Q 5.3)

the equality between the MGF of sum of Matrices and the product of their MGF doesn't hold unless they Commute. we'll show this by providing a

Counter example: $M_{(A+B)}(\theta) \stackrel{?}{=} \prod M_A$

$$\rightarrow A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\rightarrow AB = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, BA = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \rightarrow AB \neq BA$$

(they don't Commute)

$$\rightarrow e^{A+B} \approx \begin{pmatrix} 3.54 & 1.36 \\ 1.36 & 0.81 \end{pmatrix}, e^A e^B = \begin{pmatrix} 4.10 & 3.19 \\ 2.43 & 0.566 \end{pmatrix}$$

$$\rightarrow M_{A+B}(\theta) \neq M_A(\theta) M_B(\theta) \text{ (for } \theta=1) \square.$$

5.4) for this part we will also provide a Counter example

$$A = \begin{pmatrix} -0.14 & 0.67 & -0.12 \\ 0.67 & 0.13 & -0.49 \\ -0.12 & -0.49 & 0.12 \end{pmatrix}, B = \begin{pmatrix} 0.198 & 0.36 & -0.83 \\ 0.36 & -0.83 & 0.21 \\ -0.83 & 0.21 & -0.4 \end{pmatrix}$$

$$C = \begin{pmatrix} -0.35 & -1.14 & 0.46 \\ -1.14 & -0.17 & -0.56 \\ 0.46 & -0.56 & -0.94 \end{pmatrix} \rightarrow \text{Tr}(e^{A+B+C}) = 2.96$$

$$\text{Tr}(e^A e^B e^C) = 1.32$$

$$\rightarrow \text{Tr}(e^{A+B+C}) > \text{Tr}(e^A e^B e^C)$$

□.

Q5.5)

Consider the matrices A, B from part 3.

define: $X_1 = \varepsilon_1 A$, $X_2 = \varepsilon_2 A$ ($\varepsilon_1, \varepsilon_2 \sim \text{Rademacher}$)

$$\rightarrow \mathbb{E}[e^{\theta X_1}] = \frac{e^{\theta A} + e^{-\theta A}}{2} = \cosh(\theta) I \quad \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = A \right)$$

$$\text{similarly } \mathbb{E}[e^{\theta X_2}] = \cosh(\theta) I \quad \left(B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right)$$

$$\rightarrow \log \mathbb{E}[e^{\theta X_1}] + \log \mathbb{E}[e^{\theta X_2}] = 2 \log \cosh(\theta)$$

$$\rightarrow X_1 + X_2 = \varepsilon_1 A + \varepsilon_2 B$$

$$\rightarrow \mathbb{E}[e^{\theta (X_1 + X_2)}] = \cosh(\sqrt{2} \theta) I$$

$$\rightarrow \log \cosh(\sqrt{2} \theta) I = \log \mathbb{E}[e^{\theta (X_1 + X_2)}] \neq \log \mathbb{E}[e^{\theta X_1}] + \log \mathbb{E}[e^{\theta X_2}] = 2 \log \cosh(\theta) I$$

Q6) first let's try and bound the MGF of A ;

set Q to be the Hermitian dilation of A :

$$Q = \mathcal{D}(A) \rightarrow \|Q\|_{op} = \|A\|_{op}$$

$$\rightarrow \mathbb{E}[e^{\lambda Q}] \leq \dots \rightarrow \text{we want to provide a bound for this}$$

Q 6) continued,

$$e^{\lambda Q} = I + \lambda Q + \sum_{k=2} \frac{\lambda^k Q^k}{k!} \quad (\text{note that } Q = GB)$$

$$= I + \lambda GB + \sum_{k=2} \frac{\lambda^k G^k B^k}{k!}$$

$$\|B\|_{op} \leq b_2$$

$$\rightarrow B^k \leq b_2^k I_{(d_1+d_2)} \\ (\text{consider } B \text{ diagonal})$$

$$\leq I + \lambda GB + \sum_{k=2} \frac{\lambda^k G^k b_2^k I}{k!}$$

take expectation

$$\rightarrow \leq I + \sum_{k=2} \frac{\lambda^k \frac{k!}{2} b_1^{k-2} b_2^k \sigma^2 I}{k!}$$

$$= I + \frac{\lambda^2 b_1^2 \sigma^2}{2} I \times \sum_{k'=0}^{\infty} (\lambda b_1 b_2)^{k'} = I + \frac{(\lambda b_2)^2 I}{2(1-b_1 b_2 |\lambda|)}$$

$$\leq \exp\left(\frac{(\lambda b_2 \sigma)^2 I}{2(1-b_1 b_2 |\lambda|)}\right)$$

$$\rightarrow \mathcal{P}\left(\left\|\frac{1}{n} \sum A_i\right\| > \delta\right) = \mathcal{P}\left(\left\|\frac{1}{n} \sum Q_i\right\|_{op} > \delta\right)$$

$$\leq 2 e^{-\lambda \delta n} \text{tr}(\mathbb{E}[e^{\lambda \sum Q_i}]) \leq 2 e^{-\lambda n \delta} \text{tr}(\exp(\sum \log \mathbb{E} e^{\lambda Q_i}))$$

$$\leq 2 \text{tr}\left(\exp\left(\frac{\lambda^2 \sum \text{Var}(Q_i)}{1-b_1 b_2 |\lambda|}\right) e^{-\lambda \delta n}\right) \quad \left(\text{Var } Q_i = b_2^2 \sigma^2 I_{(d_1+d_2)}\right)$$

$$\leq 2 e^{-\lambda \delta n} \text{rank}\left(\sum_{d_1+d_2} \text{var}(Q_i)\right) e^{\frac{n \lambda^2 b_2^2 \sigma^2}{1-b_1 b_2 |\lambda|}}$$

Q 6) continued:

minimizing the bound achieve w.r.t. λ gives $\lambda = \frac{\delta}{b_1^2 \sigma^2 + b_1 b_2 \delta}$

$$\rightarrow P\left(\left\|\frac{1}{n} \sum Q_i\right\|_2 > \delta\right) \leq 2(d_1 + d_2) \exp\left(\frac{-n \delta^2}{2(\sigma^2 b_2^2 + b_1 b_2 \delta)}\right)$$

Q7) first we'll prove a tail bound and then use the expectation integral formula to obtain our desired bound:

$$\mathbb{E}\left[e^{\lambda \varepsilon_i A_i}\right] = \frac{e^{\lambda A} + e^{-\lambda A}}{2} = \cosh(\lambda A) \leq e^{\frac{\lambda^2 A^2}{2}}$$

the last inequality comes from $\cosh(x) \leq \exp(x^2/2)$ $x \in \mathbb{R}$ and is valid for matrices because it can be proved by comparing the Taylor expansions.

therefore $\varepsilon_i A_i$ is subgaussian with parameter A_i^2

now that we have established that, we'll move on to establishing a tail bound:

$$P\left(\left\|\sum \varepsilon_i A_i\right\| > \delta\right) \leq 2 e^{-\lambda \delta} \text{trace}\left\{\exp\left(\sum \log \varphi_{\varepsilon_i A_i}(\lambda)\right)\right\}$$

$$\leq 2 e^{-\lambda \delta} \times d e^{\frac{\lambda^2 \sigma^2}{2}} \xrightarrow[\text{minimize on } \lambda]{} 2 d \exp\left(\frac{-\delta^2}{2 \sigma^2}\right)$$

$$\sigma^2 = \left\|\sum A_i^2\right\|_{\text{op}}$$

Q7) continued: now we'll bound the expectation:

$$\mathbb{E}[\|\sum \varepsilon_i A_i\|^p] = \int_0^\infty \mathcal{P}(\|\sum \varepsilon_i A_i\|^p > s) ds$$

$$\leq \int_0^\infty 2d \exp\left(-\frac{s^{2/p}}{2\sigma^2}\right) ds$$

$$\rightarrow x^p = s \rightarrow 2d p \int x^{p-1} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx \quad \left. \begin{array}{l} \text{we'll approximate} \\ \text{this integral by} \\ \text{a gaussian moment} \end{array} \right\}$$

$$\leq 2d p \sigma^p \frac{2^{p/2} \Gamma(\frac{p+1}{2})}{\sqrt{\pi}}, \quad \Gamma\left(1 + \frac{p}{2}\right) \approx \sqrt{2\pi \frac{p}{2}} \left(\frac{p}{2e}\right)^{p/2}$$

$$\rightarrow \lesssim d \sigma^p p^{p/2} \rightarrow \text{take } p\text{-th root}$$

$$\rightarrow \lesssim d^{1/p} \sigma p^{1/2}, \quad d^{1/p} = e^{\frac{\log d}{p}} \approx 1 + \frac{\log d}{p}$$

(because $\frac{\log d}{p}$ is small this is a valid approximation.)

$$\lesssim \sigma \left(\sqrt{p} + \frac{\log d}{\sqrt{p}} \right) \leq 2\sigma \left(\sqrt{p + \log d} \right)$$

$$\rightarrow \left(\mathbb{E}[\|\sum \varepsilon_i A_i\|^p] \right)^{1/p} \leq C \sqrt{p + \log d} \left\| \sum A_i^2 \right\|_{\text{op}}^{1/2}$$

for some universal constant C .

□.

Q8) 8.1) define the distance function (error)

as follows:
$$\mathcal{D}(s) = \sup_{f \in \mathcal{G}} \mathbb{E}_z[f(z)] - \frac{1}{n} \sum f(z_i)$$

$$s = (z_1, \dots, z_n)$$

if we show that $P(\mathcal{D}(s) > 2R_n(\mathcal{G}) + \sqrt{\frac{\log 1/\delta}{2n}}) < \delta$
then the desired inequality (stated in 8.1) holds
for all $f \in \mathcal{G}$.

let's analyze $\mathcal{D}(s)$ for a fixed f .

$$\mathcal{D}_f(s) - \mathcal{D}_f(s') = \frac{1}{n} (f(z_i) - f(z'_i)) \leq \frac{1}{n}$$

$(s' = (z_1, \dots, z'_i, \dots, z_n))$

therefore $\mathcal{D}$ has the bounded difference property

with $c_i = \frac{1}{n} \rightarrow \mathcal{D}(s)$ is subgaussian with $\sigma^2 = \sum c_i^2 = \frac{1}{n}$

$$\rightarrow P(\mathcal{D}(s) \gg \mathbb{E}[\mathcal{D}(s)] + t) \leq e^{-2nt^2}$$

$$\rightarrow P(\mathcal{D}(s) \gg \mathbb{E}[\mathcal{D}(s)] + \sqrt{\frac{\log 1/\delta}{2n}}) \leq \delta$$

therefore it suffices to show that $\mathbb{E}[\mathcal{D}(s)] \leq 2R_n(\mathcal{G})$

this is done through the symmetrization argument

$$\mathbb{E}[\mathcal{D}(s)] = \mathbb{E}\left[\sup_{f \in \mathcal{G}} \mathbb{E}_z[f(z)] - \frac{1}{n} \sum f(z_i)\right] = \mathbb{E}\left[\sup_{f \in \mathcal{G}} \mathbb{E}_s \left[\frac{f(z_i) - f(z'_i)}{n} \right]\right]$$

define a ghost sample: $s' = (z_1, \dots, z'_n)$

Q8.1) continued:

$$\rightarrow \mathbb{E}_{s, s'} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(y_i) - f(z_i)) \right] = \mathbb{E}_{s, s', \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(y_i) - f(z_i)) \right]$$

(Jensen (since sup is convex)) (since our distribution is symmetric)

$$\leq 2 \mathbb{E}_{s, \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(z_i) \right] = R_n(\mathcal{G}) \quad (\text{because } \sup_{f \in \mathcal{F}} (f(y) - f(z)) \leq \sup_{f \in \mathcal{F}} f(y) + \sup_{f \in \mathcal{F}} f(z))$$

this completes our argument

Q8.2) based on the definition of the class $\mathcal{G}$

the following relation between h , γ_h is derived:

$2 \gamma_h(x, y) - 1 = h(x)$, based on this relation, we can compare the rademacher complexity.

$$\begin{aligned} R_n(\mathcal{H}) &= \mathbb{E}_{s, \varepsilon} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(x_i) \right] = \mathbb{E}_{s, \varepsilon} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (2 \gamma_h(x_i, y_i) - 1) \right] \\ &= \mathbb{E}_{s, \varepsilon} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i 2 \gamma_h(x_i, y_i) \right] - \mathbb{E}_{s, \varepsilon} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n -\varepsilon_i \right] \\ &= 2 R_n(\mathcal{G}) \quad \square. \end{aligned}$$

Q8.3) the growth function is defined as followed

$$\Pi_H(m) := \max_{(x_1, \dots, x_m)} |\{ (h(x_1), \dots, h(x_m)) : h \in \mathcal{H} \}|$$

define $V_s := \{ (h(x_1), \dots, h(x_m)) \in \{-1, 1\}^m : h \in \mathcal{H} \}$

Q8.3) continued:

clearly $|V_s| \leq \Pi_H(m)$, massart lemma states that:

$$\hat{R}_s(H) \leq \sqrt{\frac{2 \log |V_s|}{m}} \leq \sqrt{\frac{2 \log \Pi_H(m)}{m}}$$

(this is provable via the symmetrization argument that was discussed in Class.)

Q8.4)auer-Sheelah lemma states that:

$$\Pi_H(m) \leq \sum_{k=0}^d \binom{m}{k}, \quad d = VC(H)$$

$$\left(\frac{d}{m}\right)^d \sum_{k=0}^d \binom{m}{k} \leq \sum_{k=0}^d \binom{m}{k} \left(\frac{d}{m}\right)^k \leq \left(1 + \frac{d}{m}\right)^m \leq e^d$$

$$\rightarrow \sum_{k=0}^d \binom{m}{k} \leq \left(\frac{em}{d}\right)^d. \quad \square.$$

Q8.4) From part 1 we have:

$$\mathbb{E}_{Z \sim \mathcal{D}}[g(z)] \leq \frac{1}{m} \sum g(z_i) + 2 R_m(G) + \sqrt{\frac{\log 1/\delta}{2m}}$$

with probability $> 1 - \delta$

$$\text{set } R(h) = \mathbb{P}(h(x) \neq y) = \mathbb{E}_s[1(h(x) \neq y)]$$

$$, \hat{R}_s(h) = \frac{1}{m} \sum 1(h(x_i) \neq y_i)$$

$$\rightarrow 2 R_m(G) = R_m(H) \leq \sqrt{\frac{2d \log(\frac{em}{d})}{m}} \approx C_1 \sqrt{\frac{\log m/d}{m/d}}$$

$$\rightarrow R(h) \leq \hat{R}_s(h) + C_1 \sqrt{\frac{\log m/d}{m/d}} + C_2 \sqrt{\frac{1/\delta}{m}}$$

□.